1) Distance Vector routing allows nodes to share routing information with their neighbors and choose the shortest path based on hop count. Link State routing gives nodes a map of the network and uses algorithms like Dijkstra's to find shortest paths.
2) A node must accept worse routes from its current next hop to avoid using outdated information. For example, if node 2 tells node 1 that a route is no longer available, node 1 must update its routing table instead of continuing to use the broken path. If node 1 only accepted better routing information, it would continue believing that the old route was valid and keep sending packets toward node 2.
3) Split horizon prevents a node from advertising a route back to the neighbor it learned it from. Poison reverse advertises that route with an infinite cost of 16 instead. Route poisoning announces that a route has become unreachable.
4) No, nodes calculate their routes independently, so packets traveling between two nodes may take different paths in each direction.
5) If an advertisement is lost, the node can receive updated information during the next routing period, which is every three seconds. If after 12 seconds without any updates, learned routes become expired.
6) If a node repeatedly advertises and withdraws a route, or advertises a route but never forwards packets it would cause a change in routes and will lead to packet loss or delays.